Making a Fair Test

A fair test changes only one thing. Everything else stays the same.

Name ______________________ Date ______________

Worked example
Fair test: Does a paper plane fly farther with a paper clip on it?
Change: the paper clip. Measure: how far it flies.
Keep the same: the same plane, the same throw, the same starting spot.

My investigation question

The one thing I change

I only change this.

The thing I watch or measure

This is what I find out.

The things I keep the same

List everything that must not change.

Variables and Controls

Identify every variable before you start. Uncontrolled variables are the usual reason a result cannot be trusted.

Name ______________________ Date ______________

Worked example
Question: How does salt concentration affect the melting time of ice?
Independent: salt concentration (0, 5, 10, 15 g per 100 ml).
Dependent: time to melt (s), measured with a stopwatch.
Controlled: ice mass, starting temperature, container, room temperature.
Control group: ice in pure water (0 g of salt).

My investigation question

Independent variable

What you change, and the range of values you will use.

Dependent variable

What you measure, with units and the instrument used.

Controlled variables

What you hold constant, and how you will hold it constant.

Is my test fair?

If someone changed something else as well, could you still trust the result?

Variables in an Experiment

Change one variable, measure one variable, control the rest. That is what makes a test fair.

Name ______________________ Date ______________

Worked example
Question: Does ramp height change how far a toy car rolls?
Independent: ramp height (cm).
Dependent: distance rolled (cm), measured with a ruler.
Controlled: same car, same ramp surface, same starting line.

My investigation question

Independent variable

The one thing you change on purpose.

Dependent variable

The thing you measure. Give the units.

Controlled variables

Everything you deliberately keep the same.

Is my test fair?

If someone changed something else as well, could you still trust the result?